1
2
3

Question 2 Max. score: 50.00

K-XOR tree

Given an undirected tree with N nodes such that there exists a path between any two nodes. For the i^{th} node, a positive integer value A_i is associated with it.

You also have a special positive integer K and you may perform the following operation zero or more times on the given tree:

- Choose an edge and update the value of the nodes connected with that edge with Bitwise XOR of its current value and K. Formally, $A_i = (A_i \ XOR \ K)$ for all i connected with the chosen edge.

Find the maximum value of $\sum_{i=1}^{i=N} A_i$ if you can perform the above operations any number of times.

Function description

Complete the *FindMaximum* function. This function takes the following 4 parameters and returns the maximum value of the sum after performing the operations:

- *N*: Represents the number of nodes in the tree
- *edges*: Represents the edges in the given tree
- *A*: Represents the values associated with each node
- *K*: Represents the integer as described above

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line contains N which denotes the number of nodes in the tree.
- The next N - *1* line contains two spaced integers denoting an edge in the tree.
- The next line contains N space-separated integers i^{th} integer representing the value of node i.
- The next line contains a single integer which denotes K.

Output format

Print in a single line the maximum value of the given expression (given above) after performing the operations.

Constraints

$2 \leq N \leq 2 * 10^5$
$1 \leq A_i, K \leq 10^9$
$1 \leq u, v \leq N, u \neq v$

New Submission All Submissions

Auto-complete ready! Save C++17 (g++ 10.3.0)

```
#include<bits/stdc++.h>
using namespace std;

long long FindMaximum (int N, vector<vector<int> > edges, vector<int> A, int K) {
    // Write your code here

}
    cin >> K;

    long long out_;
    out_ = FindMaximum(N, edges, A, K);
    cout << out_;
}
```

1:1 vscode

Test against custom input
Custom input populated

Compile & Test code Submit code

- The next N - 1 line contains two spaced integers denoting an edge in the tree.
- The next line contains N space-separated integers i^{th} integer representing the value of node i.
- The next line contains a single integer which denotes K.

Output format

Print in a single line the maximum value of the given expression (given above) after performing the operations.

Constraints

$2 \le N \le 2 * 10^5$
$1 \le A_i, K \le 10^9$
$1 \le u, v \le N, u \ne v$

Sample input

```
3
1 2
1 3
1 1 2
3
```

Sample output

```
6
```

Explanation

You can perform the operation on edge $(1, 2)$ to get $A_1 = (1\ XOR\ 3) = 2, A_2 = (1\ XOR\ 3) = 2, A_3 = 2$.

It can be proved that this operation gives maximum $\sum_{i=1}^{i=N} A_i = A_1 + A_2 + A_3 = 2 + 2 + 2 = 6$.

The following test cases are the actual test cases of this question that may be used to evaluate your submission.

Sample input 1

```
20
1 2
2 3
3 4
4 5
5 6
6 7
7 8
8 9
9 10
```

Sample output 1

```
98007
```

New Submission   All Submissions

Auto-complete ready!   Save   C++17 (g++ 10.3.0)

```
#include<bits/stdc++.h>
using namespace std;

long long FindMaximum (int N, vector<vector<int> > edges, vector<int> A, int K) {
    // Write your code here

}
    cin >> K;

    long long out_;
    out_ = FindMaximum(N, edges, A, K);
    cout << out_;
}
```

0 / 12 Completed

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ayushk2k23@gmail.com

End test

Question 11

Max. score: 50.00

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Meeting room

Given *N* groups of people who want to have a meeting in the only meeting room in the office. Each group has *people[i]* people, who want to start the meeting at the *starting[i]* time and end it at the *ending[i]* time (both inclusive).

No two groups can use the meeting room at the same time. If group *j* does not get the meeting room, then all the *people[j]* people cannot meet.

Calculate the minimum number of people that cannot meet if the meeting room is optimally assigned to groups.

Note: Group *i* and *j* can get the meeting room one after the other if and only if *end[i] < start[j]*.

Function description

Complete the function *solve*. This function takes the following *4* parameters and returns the required answer:

- *N*: Represents the number of groups
- *people[]*: Represents the number of people in each

‹ Previous Question

Next Question ›

C (gcc 10.3)

Auto-complete ready!

Autosaved

```c
#include<stdio.h>...
long long solve (int N, int* people, int* starting, int* ending) {
    // Write your code here
}...
```

Ln 1, Col 1, Visual Studio

View test cases

Run code

1
2
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4
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12

- *N:* Represents the number of groups
- *people[]:* Represents the number of people in each group
- *starting[]:* Represents the starting time of their meetings
- *ending[]:* Represents the ending time of their meetings

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line contains *N* denoting the number of groups.
- The second line contains *N* space-separated integers denoting the number of people in each group.
- The third line contains *N* space-separated integers denoting the starting time of each group's meeting.
- The fourth line contains *N* space-separated integers denoting the ending time of each group's meeting.

Output format

C (gcc 10.3)
Auto-complete ready!
Autosaved

```
#include<stdio.h>
long long solve (int N, int* people, int* starting, int* ending) {
    // Write your code here
}
```

Ln 1, Col 1, Visual Studio

Run code

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Output format

Print a single integer representing the minimum number of people who cannot have a meeting.

Constraints

$1 \leq N \leq 10^5$
$2 \leq people[i] \leq 10^9$
$1 \leq start[i] \leq 10^9$
$1 \leq end[i] \leq 10^9$
$start[i] \leq end[i]$

Sample input

```
5
4 3 5 6 10
1 2 3 6 5
1 2 5 7 7
```

Sample output

```
10
```

Explanation

Given

- *N = 5*
- *people[] = [4, 3, 5, 6, 10]*

C (gcc 10.3)
Auto-complete ready!
Autosaved

```
#include<stdio.h>...
long long solve (int N, int* people, int* starting, int* ending) {
    // Write your code here
}...
```

Ln 1, Col 1, Visual Studio

Run code

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